

File System Mounting in Linux

Mount and Unmount Demonstration using WSL2

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Fundamentals of Operating System Design (FOSD)

Learning Objectives

- Understand the purpose of `mount` and `umount`.
- Create a filesystem inside a regular file.
- Use a loop device to mount a filesystem image.
- Verify a mount using `findmnt` and `df`.
- Create and access files inside the mounted filesystem.
- Safely unmount the filesystem.

What is Mounting?

Mount

Mounting attaches a filesystem to an existing directory in the Linux directory hierarchy.

- The directory is called the **mount point**.
- The filesystem contents become accessible through that directory.
- Linux can mount filesystems from disks, partitions, network resources, or filesystem image files.

This Demonstration

We create an ext4 filesystem inside a 100 MiB image file and mount it at ~/myfs.

Overall Demonstration

filesystem.img



Loop device



/dev/loop0



mount



~/myfs

Key Idea

A regular file is used as a container for an ext4 filesystem. The loop device makes that file

Step 1: Create a Filesystem Image

```
dd if=/dev/zero of=filesystem.img bs=1M count=100
```

- dd: low-level copy/conversion command.
- if=/dev/zero: input stream producing zero bytes.
- of=filesystem.img: output file.
- bs=1M: block size of 1 MiB.
- count=100: copy 100 blocks.

$$100 \times 1 \text{ MiB} = 100 \text{ MiB}$$

Important

The result is only a zero-filled file. It is not yet an ext4 filesystem.

Step 2: Create an ext4 Filesystem

```
mkfs.ext4 filesystem.img
```

- `mkfs`: make filesystem.
- `ext4`: filesystem type.
- Creates filesystem metadata inside the image.

Structures Created

- Superblock
- Block allocation structures
- Inodes
- Root directory

Step 3: Create a Mount Point

```
mkdir ~/myfs
```

Before mounting:

`~/myfs` $\longrightarrow$ ordinary directory

After mounting:

`~/myfs` $\longrightarrow$ root of the mounted ext4 filesystem

Mount Point

An existing directory at which a filesystem is attached.

Step 4: Mount the Filesystem Image

```
sudo mount -o loop filesystem.img ~/myfs
```

- `sudo`: run with root privileges.
- `mount`: attach the filesystem to the directory tree.
- `-o loop`: use a loop device for the image file.
- `filesystem.img`: file containing the ext4 filesystem.
- `~/myfs`: mount point.

Conceptual Operation

`filesystem.img` → **loop device** → `ext4` → `~/myfs`

What Does -o loop Mean?

A normal filesystem may be stored on a block device:

`/dev/sda1 → mount → /mnt/data`

For an image file:

`filesystem.img → loop device → mount → /mnt/data`

Loop Device

A loop device allows a regular file to be accessed as if it were a block storage device.

Step 5: Verify the Mount

```
findmnt ~/myfs
```

Typical information:

```
TARGET SOURCE FSTYPE OPTIONS  
/home/pmj/myfs /dev/loop0 ext4 rw,relatime
```

- **TARGET**: mount point.
- **SOURCE**: device providing the filesystem.
- **FSTYPE**: filesystem type.
- **OPTIONS**: mount options.

Step 6: Create a File

```
echo "Hello File System" > ~/myfs/test.txt  
ls -l ~/myfs
```

Observation

`test.txt` is stored inside the ext4 filesystem contained in `filesystem.img`.

The image file acts as the storage container for the mounted filesystem.

Step 7: Observe Disk Usage

```
df -Th ~/myfs
```

Field	Meaning
Filesystem	Device/filesystem source
Type	Filesystem type
Size	Total capacity
Used	Space currently used
Avail	Available space
Mounted on	Mount point

Teaching point: `df` reports filesystem-level storage usage.

Step 8: Unmount the Filesystem

```
sudo umount ~/myfs
```

- `umount` detaches the filesystem from the directory tree.
- The image file is not deleted.
- Data remains in `filesystem.img`.

After Unmounting

`~/myfs` becomes an ordinary directory again.

Important

The command is `umount`, not `unmount`.

Complete Demonstration

```
dd if=/dev/zero of=filesystem.img bs=1M count=100
```

```
mkfs.ext4 filesystem.img
```

```
mkdir ~/myfs
```

```
sudo mount -o loop filesystem.img ~/myfs
```

```
findmnt ~/myfs
```

```
echo "Hello File System" > ~/myfs/test.txt
```

```
ls -l ~/myfs
```

```
df -Th ~/myfs
```

```
sudo umount ~/myfs
```

Useful Verification Commands

```
mount
```

Shows currently mounted filesystems.

```
findmnt
```

Displays mounted filesystems in a structured form.

```
losetup -a
```

Shows loop devices currently associated with files.

```
lsblk -f
```

Shows block devices and filesystem information.

WSL2: Windows Drives

Windows drives are normally exposed in WSL2 under:

```
/mnt/c  
/mnt/d
```

For example:

```
cd /mnt/c  
ls
```

Important Distinction

`/mnt/c` is provided through WSL's Windows-drive integration. The loop-mounted image demonstrates the Linux filesystem mounting mechanism more directly.

Optional: Bind Mount

A bind mount exposes an existing directory at another location.

```
mkdir ~/source ~/mountpoint  
  
echo "File System Mount Demo" > ~/source/test.txt  
  
sudo mount --bind ~/source ~/mountpoint  
  
findmnt ~/mountpoint  
  
ls ~/mountpoint  
  
sudo umount ~/mountpoint
```

Difference

Loop mount: mounts a filesystem stored inside an image file.

Bind mount: exposes an existing directory at another path.

Common Problem: Target is Busy

If:

```
sudo umount ~/myfs
```

returns:

```
target is busy
```

First leave the directory:

```
cd ~  
sudo umount ~/myfs
```

If required, identify processes:

```
fuser -vm ~/myfs
```

Operating System Concepts Illustrated

- **File system:** organizes persistent data.
- **Filesystem image:** contains filesystem structures and data.
- **Block device abstraction:** loop device provides block-level access.
- **Mount point:** integrates a filesystem into the directory hierarchy.
- **VFS:** applications use a common filesystem interface.
- **Unmount:** removes the filesystem from the namespace.

Key Takeaways

- 1 `mount` attaches a filesystem to the directory hierarchy.
- 2 A mount point is an existing directory used as the access point.
- 3 `-o loop` allows a filesystem image file to be mounted.
- 4 `findmnt` helps inspect mount relationships.
- 5 `df -Th` shows filesystem type and storage usage.
- 6 `umount` detaches the filesystem.
- 7 WSL2 can demonstrate Linux mounting without a separate physical disk.

Classroom Activity

Ask students to:

- 1 Create a 50 MiB filesystem image.
- 2 Format it using ext4.
- 3 Mount it using a loop device.
- 4 Create two directories and three files.
- 5 Verify the mount using `findmnt`.
- 6 Check usage using `df -Th`.
- 7 Unmount it.
- 8 Mount it again and verify that the files still exist.

Question

Why do the files still exist after `umount`?

- Silberschatz, Galvin, Gagne, *Operating System Concepts*, 10th Edition.
- William Stallings, *Operating Systems: Internals and Design Principles*, 9th Edition.
- Linux manual pages: `mount`, `umount`, `findmnt`, `losetup`.